

OSSP-SKILL-WEEK3

Session5:

Task1: To Apply Single Quotes, Preserve Literal Content, Ignore Variable Expansion, Store Quoted Strings, Validate Parsing Results, Test Edge Cases.

Source code:

```
GNU nano 7.2

#include <stdio.h>
#include <string.h>

#define MAX_INPUT 200
#define MAX_QUOTES 20

int main() {
    char input[MAX_INPUT];
    char quoted_strings[MAX_QUOTES][MAX_INPUT];
    int quote_count = 0;

    printf("Session 5 - Single Quote Parsing\n");
    printf("Enter a command with single quotes:\n");
    printf("myshell> ");

    fgets(input, sizeof(input), stdin);
    input[strcspn(input, "\n")] = '\0';

    int i = 0;

    while (input[i] != '\0') {

        if (input[i] == '\\') {
            i++;

            int j = 0;

            while (input[i] != '\0' && input[i] != '\\') {
                if (j < MAX_INPUT - 1) {
                    quoted_strings[quote_count][j++] = input[i];
                }
                i++;
            }

            quoted_strings[quote_count][j] = '\0';

            quote_count++;
            i++;
        } else {
            i++;
        }
    }

    printf("\nQuoted strings found:\n");

    if (quote_count == 0) {
        printf("No single-quoted strings found.\n");
    } else {
        for (int k = 0; k < quote_count; k++) {
            printf("Quoted String %d: [%s]\n",
                k + 1, quoted_strings[k]);
        }
    }

    printf("\nParsing validation:\n");

    if (quote_count > 0)
        printf("Single-quote parsing successful.\n");
    else
        printf("No quoted content to validate.\n");

    return 0;
}
```

```
GNU nano 7.2

        quoted_strings[quote_count][j] = '\0';

        if (input[i] == '\\') {
            quote_count++;
            i++;
        } else {
            printf("Error: Unclosed single quote.\n");
            return 1;
        }
    } else {
        i++;
    }
}

printf("\nQuoted strings found:\n");

if (quote_count == 0) {
    printf("No single-quoted strings found.\n");
} else {
    for (int k = 0; k < quote_count; k++) {
        printf("Quoted String %d: [%s]\n",
            k + 1, quoted_strings[k]);
    }
}

printf("\nParsing validation:\n");

if (quote_count > 0)
    printf("Single-quote parsing successful.\n");
else
    printf("No quoted content to validate.\n");

return 0;
}
```

Output:

```
gopinath@GOPINATH:~$ nano session5_task1.c
gopinath@GOPINATH:~$ gcc session5_task1.c -o session5_task1
gopinath@GOPINATH:~$ ./session5_task1
Session 5 - Single Quote Parsing
Enter a command with single quotes:
myshell> echo 'Hello $USER'

Quoted strings found:
Quoted String 1: [Hello $USER]

Parsing validation:
Single-quote parsing successful.
gopinath@GOPINATH:~$
```

Observation: I learned how to process single-quoted strings and preserve their contents as literal text. I understood that variable expansion should not occur inside single quotes. I also learned how to store quoted strings, validate the parsing result, and handle errors such as an unclosed single quote.

Task2: To Apply Double Quotes, Preserve Spaces, Allow Variable Expansion, Parse Nested Tokens, Validate Outputs, Test Quoted Commands.

Source code:


```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <ctype.h>

#define MAX_INPUT 300
#define MAX_TOKENS 50
#define MAX_TOKEN_LENGTH 100

void expand_variable(char *input, char *output) {
    int i = 0;
    int j = 0;

    while (input[i] != '\0' && j < MAX_INPUT - 1) {

        if (input[i] == '$') {
            i++;

            char variable[50];
            int k = 0;

            while (isalnum((unsigned char)input[i]) || input[i] == '_') {
                if (k < 49)
                    variable[k++] = input[i];

                i++;
            }

            variable[k] = '\0';

            char *value = getenv(variable);

            if (value != NULL) {
                int v = 0;

                while (value[v] != '\0' && j < MAX_INPUT - 1) {
                    output[j++] = value[v++];
                }
            } else {
                output[j++] = input[i++];
            }
        }

        output[j] = '\0';
    }
}

int main() {
    char input[MAX_INPUT];
    char expanded[MAX_INPUT];
    char tokens[MAX_TOKENS][MAX_TOKEN_LENGTH];
```

```
^G Help      ^O Write Out  ^W Where Is   ^K Cut
^X Exit      ^R Read File  ^\ Replace    ^U Paste
```

```
Execute
Justify
```

```
GNU nano 7.2

int token_count = 0;
int i = 0;

printf("Session 5 - Double Quote Parsing\n");
printf("Double quotes preserve spaces and allow variable expansion.\n");
printf("Enter a command:\n");
printf("myshell> ");

fgets(input, sizeof(input), stdin);
input[strcspn(input, "\n")] = '\0';

if (strlen(input) == 0) {
    printf("Empty command.\n");
    return 0;
}

/* Parse input */
while (input[i] != '\0' && token_count < MAX_TOKENS) {

    while (isspace((unsigned char)input[i])) {
        i++;
    }

    if (input[i] == '\0')
        break;

    int j = 0;

    /* Double-quoted string */
    if (input[i] == '"') {
        i++;

        while (input[i] != '\0' && input[i] != '"') {
            if (j < MAX_TOKEN_LENGTH - 1)
                tokens[token_count][j++] = input[i];

            i++;
        }

        if (input[i] == '"') {
            i++;
        } else {

^G Help      ^O Write Out  ^W Where Is   ^K Cut
^X Exit      ^R Read File  ^\ Replace    ^U Paste      Execute
                                           Justify

    } else {
        printf("Error: Unclosed double quote.\n");
        return 1;
    }

    tokens[token_count][j] = '\0';
    token_count++;
}

/* Normal token */
else {
    while (input[i] != '\0' &&
           !isspace((unsigned char)input[i]) &&
           input[i] != '"') {

        if (j < MAX_TOKEN_LENGTH - 1)
            tokens[token_count][j++] = input[i];

        i++;
    }

    tokens[token_count][j] = '\0';

    if (j > 0)
        token_count++;
}
}

printf("\nTokens before variable expansion:\n");

for (int k = 0; k < token_count; k++) {
    printf("Token %d: [%s]\n", k + 1, tokens[k]);
}

printf("\nVariable expansion:\n");

for (int k = 0; k < token_count; k++) {

    expand_variable(tokens[k], expanded);

    printf("Token %d: [%s]\n", k + 1, expanded);
}

printf("\nParsing validation:\n");
printf("Double-quote parsing successful.\n");
printf("Spaces inside quotes were preserved.\n");
printf("Variable expansion was processed.\n");

return 0;
}
```

```
GNU nano 7.2

    } else {
        printf("Error: Unclosed double quote.\n");
        return 1;
    }

    tokens[token_count][j] = '\0';
    token_count++;
}

/* Normal token */
else {
    while (input[i] != '\0' &&
           !isspace((unsigned char)input[i]) &&
           input[i] != '"') {

        if (j < MAX_TOKEN_LENGTH - 1)
            tokens[token_count][j++] = input[i];

        i++;
    }

    tokens[token_count][j] = '\0';

    if (j > 0)
        token_count++;
}
}

printf("\nTokens before variable expansion:\n");

for (int k = 0; k < token_count; k++) {
    printf("Token %d: [%s]\n", k + 1, tokens[k]);
}

printf("\nVariable expansion:\n");

for (int k = 0; k < token_count; k++) {

    expand_variable(tokens[k], expanded);

    printf("Token %d: [%s]\n", k + 1, expanded);
}

^G Help      ^O Write Out  ^W Where Is   ^K Cut
^X Exit      ^R Read File  ^\ Replace    ^U Paste
```

```
printf("\nParsing validation:\n");
printf("Double-quote parsing successful.\n");
printf("Spaces inside quotes were preserved.\n");
printf("Variable expansion was processed.\n");

return 0;
}
```

Output:

```
gopinath@GOPINATH:~$ gcc session5_task2.c -o session5_task2
gopinath@GOPINATH:~$ ./session5_task2
Session 5 - Double Quote Parsing
Double quotes preserve spaces and allow variable expansion.
Enter a command:
myshell> echo "Hello $USER"

Tokens before variable expansion:
Token 1: [echo]
Token 2: [Hello $USER]

Variable expansion:
Token 1: [echo]
Token 2: [Hello gopinath]

Parsing validation:
```

```
Double-quote parsing successful.
Spaces inside quotes were preserved.
Variable expansion was processed.
gopinath@GOPINATH:~$
```

Observation: I learned how to process double-quoted strings while preserving spaces within them. I understood how variable expansion can be performed inside double quotes, such as expanding environment variables like \$USER. I also learned how to parse quoted and nested tokens, validate the parsing results, and detect errors such as unclosed double quotes.

Session6:

Task1: To Create Process Escape Sequences, Handle Escaped Spaces, Escape Special Symbols, Preserve Characters, Validate Parser Output, Test Complex Inputs.

Source code:

```
GNU nano 7.2

#include <stdio.h>
#include <string.h>

#define MAX_INPUT 300

int main() {
    char input[MAX_INPUT];
    char output[MAX_INPUT];

    int i = 0;
    int j = 0;
    int escaped = 0;

    printf("Session 6 - Escape Sequence Parsing\n");
    printf("Enter a command with escaped spaces or special characters:\n");
    printf("myshell> ");

    fgets(input, sizeof(input), stdin);
    input[strcspn(input, "\n")] = '\0';

    if (strlen(input) == 0) {
        printf("Empty input.\n");
        return 0;
    }

    while (input[i] != '\0' && j < MAX_INPUT - 1) {

        if (escaped) {
            output[j++] = input[i];
            escaped = 0;
        } else if (input[i] == '\\') {
            escaped = 1;
        } else {
            output[j++] = input[i];
        }

        i++;
    }

    if (escaped) {
```

```
        if (escaped) {
            printf("\nWarning: Escape character at the end of input.\n");
        }

        output[j] = '\0';

        printf("\nOriginal input:\n");
        printf("%s\n", input);

        printf("\nProcessed input:\n");
        printf("%s\n", output);

        printf("\nParser validation:\n");
        printf("Escape sequences processed successfully.\n");
        printf("Escaped characters were preserved.\n");

        return 0;
    }
}
```

Output:

```
gopinath@GOPINATH:~$ nano session6_task1.c
gopinath@GOPINATH:~$ gcc session6_task1.c -o session6_task1
gopinath@GOPINATH:~$ ./session6_task1
Session 6 - Escape Sequence Parsing
Enter a command with escaped spaces or special characters:
myshell> hello\ world

Original input:
hello\ world

Processed input:
hello world

Parser validation:
Escape sequences processed successfully.
Escaped characters were preserved.
gopinath@GOPINATH:~$
```

Observation: I learned how to handle escape sequences while processing command input. I understood how escaped spaces and special symbols can be preserved as literal characters instead of being treated as separators or operators. I also learned how to detect escape characters, process escaped input, validate parser output, and handle complex user inputs.

Task2: To Create Child Processes, Execute Programs, Handle Execution Errors, Pass Arguments, Manage Parent Process, Test Command Launching.

Source code:

```
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <stdlib.h>

int main() {
    pid_t pid;
    int status;

    printf("Session 6 - Process Execution\n");

    pid = fork();

    if (pid < 0) {
        perror("fork failed");
        return 1;
    }

    if (pid == 0) {
        /* Child process */
        printf("\nChild Process\n");
        printf("Child PID: %d\n", getpid());
        printf("Parent PID: %d\n", getppid());

        printf("Executing: ls -l\n");

        execlp("ls", "ls", "-l", NULL);

        /* This runs only if exec fails */
        perror("exec failed");
        exit(EXIT_FAILURE);
    }
    else {
        /* Parent process */
        printf("\nParent Process\n");
        printf("Parent PID: %d\n", getpid());
        printf("Child PID: %d\n", pid);
        printf("Parent waiting for child...\n");

        waitpid(pid, &status, 0);

        if (WIFEXITED(status)) {
            printf("\nChild exited with status: %d\n",
                WEXITSTATUS(status));
        } else {
            printf("\nChild terminated abnormally.\n");
        }

        printf("Parent process completed.\n");
    }

    return 0;
}
```

^G Help ^O Write Out ^W Where Is ^K Cut
^X Exit ^R Read File ^\ Replace ^U Paste

Output:

```
Parent PID: 536
Child PID: 537
Parent waiting for child...

Child Process
Child PID: 537
Parent PID: 536
Executing: ls -l
total 332
-rw-r--r-- 1 gopinath gopinath 53 Aug 8 06:04 destination.txt
drwxr-xr-x 5 gopinath gopinath 4096 Aug 11 06:52 ossp_2520090049
-rw-r--r-- 1 gopinath gopinath 41 Aug 8 08:32 sample.txt
-rwxr-xr-x 1 gopinath gopinath 16232 Aug 11 10:15 session2_task1
-rw-r--r-- 1 gopinath gopinath 559 Aug 11 10:15 session2_task1.c
-rwxr-xr-x 1 gopinath gopinath 16104 Aug 11 14:39 session2_task2
-rw-r--r-- 1 gopinath gopinath 537 Aug 11 14:39 session2_task2.c
-rwxr-xr-x 1 gopinath gopinath 16680 Aug 11 16:22 session3_task1
-rw-r--r-- 1 gopinath gopinath 4230 Aug 11 16:22 session3_task1.c
-rwxr-xr-x 1 gopinath gopinath 16400 Aug 11 16:34 session3_task2
-rw-r--r-- 1 gopinath gopinath 2188 Aug 11 16:34 session3_task2.c
-rwxr-xr-x 1 gopinath gopinath 16192 Aug 11 17:28 session4_task1
-rw-r--r-- 1 gopinath gopinath 2266 Aug 11 17:28 session4_task1.c
-rwxr-xr-x 1 gopinath gopinath 16280 Aug 11 17:41 session4_task2
-rw-r--r-- 1 gopinath gopinath 2252 Aug 11 17:40 session4_task2.c
-rwxr-xr-x 1 gopinath gopinath 16192 Aug 12 03:09 session5_task1
-rw-r--r-- 1 gopinath gopinath 1552 Aug 12 03:09 session5_task1.c
-rwxr-xr-x 1 gopinath gopinath 16320 Aug 12 11:44 session5_task2
-rw-r--r-- 1 gopinath gopinath 3236 Aug 12 11:53 session5_task2.c
-rwxr-xr-x 1 gopinath gopinath 16192 Aug 12 11:58 session6_task1
-rw-r--r-- 1 gopinath gopinath 1222 Aug 12 11:58 session6_task1.c
-rwxr-xr-x 1 gopinath gopinath 16360 Aug 12 12:02 session6_task2
-rw-r--r-- 1 gopinath gopinath 1190 Aug 12 12:02 session6_task2.c
-rwxr-xr-x 1 gopinath gopinath 16312 Aug 11 07:10 skill1_task2
-rw-r--r-- 1 gopinath gopinath 879 Aug 11 09:58 skill1_task2.c
-rw-r--r-- 1 gopinath gopinath 53 Aug 8 06:03 source.txt
-rwxr-xr-x 1 gopinath gopinath 16528 Aug 8 04:41 week1pt1
-rw-r--r-- 1 gopinath gopinath 1350 Aug 8 04:41 week1pt1.c
-rwxr-xr-x 1 gopinath gopinath 16224 Aug 8 06:03 week2pt1
-rw-r--r-- 1 gopinath gopinath 1208 Aug 8 06:02 week2pt1.c
-rwxr-xr-x 1 gopinath gopinath 16304 Aug 8 03:26 week3pt1
-rw-r--r-- 1 gopinath gopinath 1023 Aug 8 03:26 week3pt1.c
-rwxr-xr-x 1 gopinath gopinath 16136 Aug 8 03:33 week3pt2
-rw-r--r-- 1 gopinath gopinath 376 Aug 8 03:32 week3pt2.c

Child exited with status: 0
Parent process completed.
```

Observation: I learned how to create a child process using `fork()` and execute another program using `exec()`. I understood how to pass arguments to the executed program and how the parent process can wait for the child using `waitpid()`. I also learned how to handle execution errors and check the child's exit status using process-management functions.

